About Andela

Andela exists to connect brilliance and opportunity. Since 2014, we have been dedicated to breaking down global barriers and accelerating the future of work for both technologists and organizations around the world. For technologists, Andela offers competitive long-term career opportunities with leading organizations, access to a global community of professionals, and educational opportunities with leading technology providers.

At Andela, we’re deeply passionate about creating long-lasting and transformative growth opportunities for all - and doing it in an E.P.I.C. way! We’re excited to continue building our remote-first team with incredible people like you. After applying for this role, you will join our Andela Community of brilliant technologists by passing a technical screening and live interview. As a community member, you’ll have access to many exclusive technologist roles. Join Andela today to access this opportunity and more in our global marketplace!

Our roles are typically filled at lightning speed, so if you’re considering applying, get your application in quickly!

Description

About

Client is a B2B marketplace for business catering. We are building a ground-up re-architecture of our platform – a headless, channel-agnostic core with AI as a cross-cutting layer.

The stack is modern and intentional: Go, PostgreSQL, Kafka (MSK), Temporal, Vespa, SpiceDB, EKS. Each business capability owns its data in a dedicated PostgreSQL instance – no shared databases.

The Role

You will design and build the production data models that become the platform's foundation.

The domain is a B2B catering marketplace with multi-level corporate customer hierarchies,restaurant partner hierarchies, multi-tenant authorization, order lifecycles spanning multiple services, and rich menu data.

The scope spans master data management, user identity and entitlements, order lifecycle, menu data, and cross-service data propagation via Kafka events. A critical design constraint:

every domain event schema serves dual consumers – operational services (search indexing,cross-service propagation) and an analytical data platform (Snowflake via Kafka Connect,Medallion architecture). Schema decisions must account for both.

This is hands-on, high-output work – the deliverable is production schemas, not slide decks.

What You Own

Domain-Driven Monolith Decomposition: Lead the data strategy to extract core catering domains (Customers, Caterers, Menus, Orders, Payments, Delivery) from the legacy monolith into service-owned databases.

Business-to-Engineering Translation: Act as the critical bridge, capturing complex business requirements and translating them from high-level conceptual models into precise, highly extensible logical and physical database structures.

Order & Menu Domain Models: Cart through checkout, complex multi-caterer menus (decoupling dishes from dynamic menus), payment, fulfillment, and delivery. Ensure transactional integrity via durable workflows across multiple service-owned databases.

Organizational Hierarchy Schemas: PostgreSQL schemas for multi-level hierarchies (customer-side and supply-side), including evaluating fixed-tier vs. self-referencing hierarchy models.

Index Strategies, Transaction Boundaries & Query Patterns: Define performance-optimized access patterns for hot paths. Own the design of ACID transaction commit boundaries, lock management for high-throughput OLTP, and data archival/offloading strategies to keep the primary database lightweight.

Cross-Service Data Patterns: UUID-based logical references between services (no cross-service physical foreign keys), event-driven propagation via Kafka with Avro schemas, write-time API validation, and periodic reconciliation.

Event Schema Design: Avro event schemas optimized for dual-consumer design (operational microservices and analytical consumers like Snowflake via Kafka Connect).

Legacy Migration: Migration scripts and data validation for moving from legacy schemas to the new platform over the 2-year timeline.

What You'll Deliver

Scalable ERDs: Conceptual, logical, and physical ERDs with full table definitions, constraints, indexes, and access pattern documentation.

Production PostgreSQL Schemas: Master data, user identity, authorization, and core catering domains (menus, orders, fulfillment).

Performance & Archival Playbooks: Documented transaction boundary definitions, lock management strategies, indexing/partitioning for high-volume tables, and automated archival patterns.

Design Decision Documentation: Trade-offs, rejected alternatives, and clear business rationales.

Avro Event Schemas: Validated for all implemented domains for both operational and analytical consumption.

Migration Scripts: Data validation for the legacy-to-new-platform schema transformation.

Data Quality Contracts: Rules per domain dictating what constraints and invariants the schema enforces.

What We Are Looking For

Required

High-Volume Transactional (OLTP) Mastery: Deep understanding of database transaction boundaries, lock management, application impact, and performance tuning for high-throughput systems.

Deep PostgreSQL Expertise: Schema design, indexing, query optimization, partitioning, JSONB, constraint design. This is your primary tool.

Business Domain Modeling & Communication: Extensive experience designing data models for complex business domains. You must be able to explain the "why" behind data modeling decisions to non-technical business leaders just as easily as you dictate engineering specs.

Database-per-Service Architecture: Designing schemas that reference data owned by other services without physical foreign keys. You understand cross-service data boundaries.

Hierarchical Data Modeling: Adjacency lists, materialized paths, closure tables. Understanding of trade-offs for shallow vs. deep hierarchies.

Event-Driven Data Propagation: Kafka or similar: how domain events carry data between services and how consumers materialize what they need.

Distinguishing

B2B SaaS / Marketplace Models: Experience modeling multi-tenant hierarchies, catering/food-tech domains, multi-vendor carts, and complex order lifecycles.

Identity & Authorization Models: Entitlements, scope-based access, relationship-based access control.

Analytical Data Modeling Awareness: Dimensional modeling, star schemas, Medallion architecture. You design with analytical consumption in mind (Snowflake/dbt).

Go Proficiency: Willingness to work in Go (migrations, data access layers, tooling).

Legacy Data Migration: Deduplication, normalization, and complex schema transformation from monolith to distributed systems.

Environment & Stack

Primary Data Store: PostgreSQL (RDS, dedicated per service)

Language: Go

Events: Amazon MSK (Kafka), Avro schemas, schema registry

Analytics (Downstream): Snowflake + dbt (Medallion architecture). Events land via Kafka Connect. You design the schemas; the Data Platform team owns the dbt transformations.

Cache / Fast tate: Redis-compatible cache

Document Patterns: PostgreSQL JSONB for versioned documents

Infrastructure: Kubernetes, Terraform

How We Measure Success

Monolith Decommissioning: Production schemas deployed for master data, identity, authorization, menus, and orders, allowing the business to decommission legacy components without requiring rework.

Comprehensive Documentation: Every schema has documented ERDs, access patterns, index strategies, transaction boundaries, and design rationales.

Cross-Service Patterns Established: UUID logical references, Kafka event propagation, write-time validation, and reconciliation are validated and working.

Repeatable Methodology: The team has a repeatable data modeling approach and can extend the patterns independently for future features.

Note: Foundational architecture work is already underway. Draft ERDs exist for core domains and cross-service data propagation patterns are established. This engagement builds on that foundation – not from scratch.

At Andela, we know our strengths lie in our diverse community whose talents, perspectives, backgrounds, and orientations we take pride in. Andela is committed to nurturing a work environment where all individuals are treated with respect and dignity. Everyone has the right to work in a professional atmosphere that promotes equal employment opportunities and prohibits discriminatory practices. Andela provides equal employment opportunities to all employees and applicants without regard to factors including but not limited to race, color, religion, gender, sexual orientation, gender identity, national origin, age, disability, pregnancy (including breastfeeding), genetic information, HIV/AIDS or any other medical status, family or parental status, marital status, amnesty or status as a covered veteran in accordance with applicable federal, state and local laws. This commitment applies to all terms and conditions of employment, including but not limited to hiring, placement, promotion, termination, layoff, recall, transfer, leaves of absence, compensation, and training. Our policies expressly prohibit any form of harassment and/or discrimination, as stated above